

# Continuous Convergence and Curried Functional Spaces

## 1. The language of filters

**Definition 1.1.** Let  $X$  be a set. A *filter* on  $X$  is a non-empty set  $F \subset 2^X$  which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on  $X$  will be denoted by  $\mathcal{F}(X)$ .

**Definition 1.2.** Let  $X$  be a set. A family  $\mathcal{B} \subset 2^X$  is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter  $[\mathcal{B}]$  generated by  $\mathcal{B}$  is defined as the collection of all supersets of sets from  $\mathcal{B}$ .

If  $\mathcal{B} = \{\{x\}\}$  where  $x \in X$ , the filter  $[x] = [\{\{x\}\}]$  is called the *universal filter* of  $x$ .

If  $f : X \rightarrow Y$  is a map and  $F \in \mathcal{F}(X)$  is a filter on  $X$ , then  $\{f(A) \mid A \in F\}$  is a filter base that generates the *image filter* denoted  $f(F)$ .

**Definition 1.3.** Let  $X$  be a topological space. Then we say that a filter  $F$  on  $X$  *converges* to a point  $x \in X$ , and write  $F \rightarrow x$ , if it contains all open neighborhoods of  $x$ .

**Remark 1.1.** The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence  $\{x_n\}_{n=1}^{\infty}$  converges to a point  $x \in X$  iff the filter

$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

converges to  $x$ . This filter  $F$  is called the *derived filter* of  $\{x_n\}_{n=1}^{\infty}$ .

- A subset  $E \subset X$  is closed if for every point  $x \in E$  there is a filter  $F \in \mathcal{F}(X)$  converging to  $x$  such that  $A \cap E \neq \emptyset$  for all  $A \in F$ .
- A topological space  $X$  is compact iff every filter  $F$  on  $X$  is contained in a convergent filter on  $X$ .
- A function  $f : X \rightarrow Y$  between topological spaces is continuous iff it preserves convergent filters, i.e.  $F \rightarrow x \in X$  implies  $f(F) \rightarrow f(x) \in Y$ .

## 2. Curried functions

**Definition 2.1.** Let  $n \in \mathbb{N}_0 = \{0, 1, 2, \dots\}$ . The set  $H_n(\mathbb{R})$  of *curried  $n$ -variable functions* is a topological vector space defined as follows:

- If  $n = 0$ , we set  $H_0(\mathbb{R}) = \mathbb{R}$ ;
- If  $H_n(\mathbb{R})$  is defined, we define  $H_{n+1}(\mathbb{R})$  to be the set of all functions from  $\mathbb{R}$  to  $H_n(\mathbb{R})$ . This set is given the pointwise topological and linear structure. In other words, a filter  $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$  converges to  $f \in H_{n+1}(\mathbb{R})$  if and only if  $F(x) = \{A(x) \mid A \in F\}$  converges to  $f(x)$  for all  $x \in \mathbb{R}$ . It is not hard to see that this topology is compatible with the pointwise vector structure.

**Definition 2.2.** Let  $\tilde{f} : \mathbb{R}^n \rightarrow \mathbb{R}$  be a function. Define

$$\gamma(\tilde{f})(x_1)(x_2)\dots(x_n) = \tilde{f}(x_1, x_2, \dots, x_n).$$

Then  $\gamma : \text{Hom}(\mathbb{R}^n, \mathbb{R}) \rightarrow H_n(\mathbb{R})$  is clearly a bijection. The function  $f = \gamma(\tilde{f})$  will be called the *curried version* of  $\tilde{f}$ .

### 3. Continuous functions

**Definition 3.1.** One may try to decouple the notion of convergent filters from topology. Let  $X$  be a set with a relation  $\rightarrow \subset \mathcal{F}(X) \times X$ . Then  $(X, \rightarrow)$  is called a *convergence space* if

- For all  $x \in X$ , we have  $[x] \rightarrow x$ ;
- If  $F \rightarrow x$  and  $G \rightarrow x$ , then  $F \cap G \rightarrow x$ ;
- If  $F \rightarrow x$  and  $F \subset G$ , then  $G \rightarrow x$ .

**Definition 3.2.** A map  $f : X \rightarrow Y$  between convergence spaces is said to be *continuous* if  $f(F) \rightarrow f(x) \in Y$  whenever  $F \rightarrow x \in X$ .

**Definition 3.3.** Let  $X, Y$  be two convergence spaces. Then the product  $X \times Y$  is endowed with the following convergence structure: a filter  $F \in \mathcal{F}(X \times Y)$  converges to  $(x, y) \in X \times Y$  if and only if  $p_X(F) \rightarrow x \in X$  and  $p_Y(F) \rightarrow y \in Y$ .

**Definition 3.4.** (see [2], pages 25-26): Let  $X, Y$  be convergence spaces. By  $\mathcal{C}(X, Y)$  denote the set of all continuous functions from  $X$  to  $Y$ . Then  $\mathcal{C}(X, Y)$  can be endowed with a convergence structure as follows. For a filter  $F \in \mathcal{F}(\mathcal{C}(X, Y))$ , we say that  $F \rightarrow f \in \mathcal{C}(X, Y)$  if and only if  $F(P) \rightarrow f(p)$  whenever  $P \rightarrow p \in X$ . Here, the filter  $F(P)$  is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on  $\mathcal{C}(X, Y)$  is called the structure of *continuous convergence*.

**Remark 3.1.** Continuous convergence on  $\mathcal{C}(X, Y)$  is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \rightarrow Y$$

continuous. Here, the product  $\mathcal{C}(X, Y) \times X$  is endowed with the product convergence structure as per [Definition 3.3](#).

**Proposition 3.1.** (see [2], page 27): *Let  $X, Y, Z$  be convergence spaces. Then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \rightarrow \mathcal{C}(X, \mathcal{C}(Y, Z))$$

*defined as  $\gamma(f)(x)(y) = f(x, y)$ , is well-defined and is a homeomorphism.*

**Proposition 3.2.** (see [2], page 34): *Let  $X, Y$  be topological spaces such that  $X$  is locally compact and  $Y$  is regular. Then the convergence space  $\mathcal{C}(X, Y)$  is topologized by the compact-open topology (see [3]). That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in [Definition 3.4](#).*

**Proposition 3.3.** (see [2], page 28): *Let  $X$  be a convergence space and let  $Y$  be a Hausdorff convergence space (i.e. every filter in  $Y$  converges to at most one point). Then  $\mathcal{C}(X, Y)$  is also a Hausdorff convergence space.*

**Proposition 3.4.** (see [2], page 29): *Let  $X$  be a convergence space and  $Y$  be a convergence vector space, i.e. a set endowed with both structures such that the operations  $+: Y \times Y \rightarrow Y$  and  $\cdot: \mathbb{R} \times Y \rightarrow Y$  are continuous. Then  $\mathcal{C}(X, Y)$  is also a convergence vector space.*

**Corollary 3.1.** *If  $X$  is a locally compact topological space and  $Y$  is a regular topological vector space, then  $\mathcal{C}(X, Y)$  is a topological vector space.*

**Proposition 3.5.** *Let  $X, Y$  be topological spaces such that  $X$  is first countable. Then a sequence  $\{f_k\}_{k \in \mathbb{N}} \subset \mathcal{C}(X, Y)$  converges to a function  $f \in \mathcal{C}(X, Y)$  (in the sense that its derived filter converges) if and only if  $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$  whenever  $x_k \xrightarrow[k \rightarrow \infty]{} x$  in  $X$ .*

*Proof:* First, let  $F$  be the derived filter of  $\{f_k\}_{k \in \mathbb{N}}$  and suppose that  $F$  converges to a function  $f \in \mathcal{C}(X, Y)$ . Consider a sequence  $\{x_k\}_{k \in \mathbb{N}}$  converging to  $x \in X$ , with its derived filter  $P$ . Now let  $V$  be a neighborhood of  $f(x)$  in  $Y$ . By the definition of continuous convergence, we have  $F(P) \rightarrow f(x)$ , and so  $V \in F(P)$ . This means that  $V$  contains an element of the base of  $F(P)$ , namely a set

$$B_{n_0, m_0} = \{f_n(x_m) \mid n \geq n_0, m \geq m_0\}.$$

Hence, for  $k \geq \max(n_0, m_0)$ , we have  $f_k(x_k) \in V$ , as desired.

Conversely, suppose that  $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$  whenever  $x_k \xrightarrow[k \rightarrow \infty]{} x \in X$ . First let us show that for any subsequence  $\{f_{n_k}\}_{k \in \mathbb{N}}$ , we have  $f_{n_k}(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ .

To this end, let  $n \in \mathbb{N}$ . We let  $z_n = x_{\varphi(n)}$ , where  $\varphi(n)$  is the minimal number of those  $k \in \mathbb{N}$  that satisfy  $n \leq n_k$ . We clearly see that  $n_1 \geq n_2$  implies  $\varphi(n_1) \geq \varphi(n_2)$  and that  $\varphi(n_k) = k$  for all  $k \in \mathbb{N}$ . Therefore, the sequence  $\{z_n\}_{n \in \mathbb{N}}$  converges to  $x$ . Hence we have  $f_k(z_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ , and so

$$f_{n_k}(x_k) = f_{n_k}(z_{n_k}) \xrightarrow[k \rightarrow \infty]{} f(x). \quad (1)$$

Now, to show that  $f_k \rightarrow f$  in  $\mathcal{C}(X, Y)$ , we need to show that the derived filter  $F$  generated by the sets  $C_n = \{f_k \mid k \geq n\}$  converges to  $f$ . To this end, let  $P$  be a filter in  $X$  converging to a point  $x \in X$ . Note that since  $X$  is first countable, the filter  $P$  has a countable base  $\{A_m\}_{m \in \mathbb{N}}$  of neighborhoods of  $x$ . To show that  $F(P)$  converges to  $f(x)$ , assume the contrary. Then some neighborhood  $V$  of  $f(x)$  does not lie in  $F(P)$ , which is to say that for all  $n, m \in \mathbb{N}$ , we have  $C_n(A_m) \not\subset V$ . That means that for all  $n, m \in \mathbb{N}$ , there are  $k_m \geq m$  and  $x_m \in A_m$  such that  $f_{k_m}(x_m) \notin V$ .

In particular, we have  $f_{k_n}(x_n) \notin V$  for all  $n \in \mathbb{N}$ . At the same time, since the sets  $A_n$  generate  $P$ , we see that  $x_n \xrightarrow[n \rightarrow \infty]{} x$ , so we must have  $f_{k_n}(x_n) \xrightarrow[n \rightarrow \infty]{} f(x)$  by (1), a contradiction. ■

**Definition 3.5.** Let  $n \in \mathbb{N}_0$ . The set  $C_n(\mathbb{R})$  of *curried continuous functions* is a Hausdorff topological vector space defined recursively as follows:

- If  $n = 0$ , we set  $C_0(\mathbb{R}) = \mathbb{R}$ .
- If  $C_n(\mathbb{R})$  is defined, we set  $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$ . Proposition 3.2, Proposition 3.3, and Corollary 3.1 ensure that  $C_{n+1}(\mathbb{R})$  is a Hausdorff topological vector space, provided that  $C_n(\mathbb{R})$  is.

**Proposition 3.6.** *Let  $n \in \mathbb{N}_0$ . Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \rightarrow C_n(\mathbb{R})$$

*defined by  $\gamma(f)(x_1)(x_2)\dots(x_n) = f(x_1, \dots, x_n)$ , is well-defined and is a homeomorphism.*

*Proof:* If  $n = 0$ , the statement is trivial. Assume it is proven for  $C_n(\mathbb{R})$ . By

Proposition 3.1, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ■

## 4. Differentiable functions

**Definition 4.1.** Let  $n \in \mathbb{N}$ . A curried function  $f \in H_n(\mathbb{R})$  is called *differentiable* at  $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$  if

- $f(x_1) \in H_{n-1}(\mathbb{R})$  is differentiable at  $(x_2, \dots, x_n) \in \mathbb{R}^{n-1}$ , if  $n \geq 2$ ;
- There exists a continuous function  $f'(x_1) \in C_{n-1}(\mathbb{R})$  such that

$$f(x_1 + h) = f(x_1) + h \cdot f'(x_1) + o(h),$$

where  $o(h) \in C_{n-1}(\mathbb{R})$  is a function such that  $\frac{o(h)}{h} \rightarrow 0$  as  $h \rightarrow 0$ . Equivalently, we may write

$$f'(x_1) = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}.$$

**Proposition 4.1.** *If  $f \in H_n(\mathbb{R})$  is differentiable everywhere, then  $f$  is continuous.*

*Proof:* If  $n = 1$ , then  $f : \mathbb{R} \rightarrow \mathbb{R}$  is differentiable in the usual sense, and so  $f$  is continuous.

Suppose that the statement is proven for  $n - 1$ , and consider a differentiable function  $f \in H_n(\mathbb{R})$ . First of all, for all  $x \in \mathbb{R}$  we know that  $f(x) \in H_{n-1}(\mathbb{R})$  is differentiable, and hence continuous. It remains to show that  $f$  is a continuous map from  $\mathbb{R} \rightarrow C_{n-1}(\mathbb{R})$ .

Let  $x_k \rightarrow x_0 \in \mathbb{R}$ . Then, letting  $h_k = x_k - x_0$ , we can write

$$f(x_k) = f(x_0 + h_k) = f(x_0) + h_k \cdot f'(x_0) + o(h_k),$$

where  $o(h_k) \in C_{n-1}(\mathbb{R})$  is such that  $o(h_k)/h_k \xrightarrow[k \rightarrow \infty]{} 0$ . Since  $f'(x_0)$  is continuous, we have

$$h_k \cdot f'(x_0) + o(h_k) \xrightarrow[k \rightarrow \infty]{} 0 \in C_{n-1}(\mathbb{R}),$$

so  $f(x_k) \rightarrow f(x_0)$ , which means that  $f$  is continuous. ■

**Proposition 4.2.** *A curried function  $f \in H_n(\mathbb{R})$  is differentiable at  $(x_1, \dots, x_n)$  if and only if its counterpart  $\tilde{f} = \gamma^{-1}(f) : \mathbb{R}^n \rightarrow \mathbb{R}$  is differentiable at  $(x_1, \dots, x_n)$ .*

*Proof:* ■

## Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

- [2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.
- [3] R. H. Fox, "On Topologies For Function Spaces," 1945.